Homework 2

Due Thursday, September 17, 2026 by 11:59 PM on Gradescope

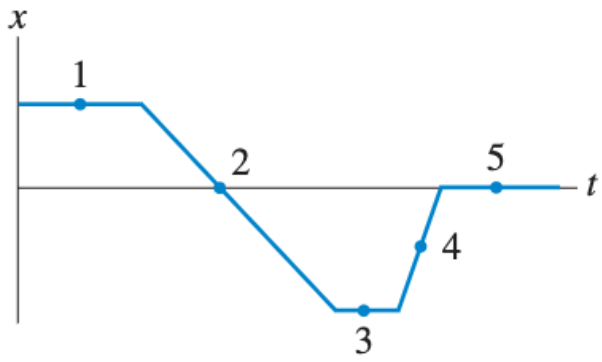
Instructions

- Include your full name and Coyote ID on the top right of the first page
- Clearly circle or box your answer for any question involving calculations. If the problem is multipart, circle or box answers for each part.
- Work should be neatly written and easy to follow.
- You must tag each question on Gradescope. If you do not tag your work for a given problem, you will not receive credit for it.

Problems

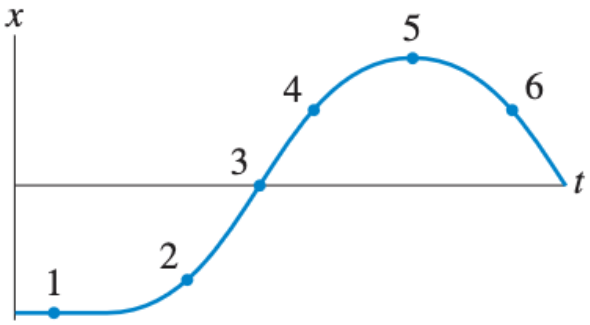
This homework contains problems related to 1D motion and 1D kinematics.

Figure 1



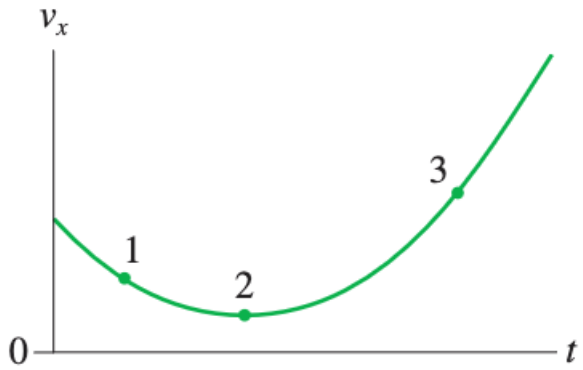
1. Figure 1 shows the position vs. time graph for a moving object. At which numbered point or points is the object:
   a. moving the slowest?
   b. moving the fastest?
   c. at rest?
   d. moving to the left?

Figure 2



2. Figure 2 shows the position vs. time graph for a moving object. At which numbered point or points is the object:
   a. moving the slowest?
   b. moving the fastest?
   c. at rest?
   d. moving to the left?
3. You're driving along the highway at a steady speed of $60\ \mathrm{mi/h}$ when another driver decides to pass you. At the moment when the front of his car is exactly even with the front of your car, and you turn your head to smile at him, do the two cars have equal velocity? Explain.
4. A bicycle is traveling east. Can its acceleration vector ever point west? Explain.

Figure 3



5. Figure 3 shows the velocity-vs.-time graph for a moving object. At which number point or points is the object:
   a. speeding up?
   b. slowing down?
   c. moving to the left?
   d. moving to the right?
6.
   a. What constant acceleration, in SI units, must a car have to go from zero to $60\,\mathrm{mi/h}$ in $10\,\mathrm{s}$?
   b. How far has the care traveled when it reaches $60\,\mathrm{mi/h}$? Give your answer both in SI units and in feet.

7. It has been proposed that a very small probe could be sent to a nearby star system by using a powerful leaser beam, fired from an earth-orbiting satellite, to push on a lightweight “solar sail.” Very high speeds could be reached in the vacuum of space by a fairly modest acceleration that continues for a long interval of time.
   a. Write an expression for the constant acceleration a_x an object needs to reach velocity $v_{\max}$ in time t_{push}, starting from rest.
   b. Write an expression in terms of $v_{\max}$ and t_{push} for the distance d the object travels during this time.
   c. For the mission to be feasible, the probe needs to reach 10% of the speed of light after being pushed for $1.0\ \text{year}$. The probe would then coast the rest of the way. What constant acceleration is needed? (Note the speed of light and other useful data needed to solve problems are given inside the root and back covers of the book)
   d. What fraction of a light year will the probe have traveled at the end of the year? A light year (ly) is the distance traveled by light in $1\ \text{year}$
8. When you sneeze, the air in your lungs accelerates from rest to $150\ \text{km/h}$ in approximately $0.50\ \text{s}$. What is the magnitude of the acceleration of the air in m/s^2?

9. Ball bearings are made by letting spherical drops of molten metal fall inside a tall tower—called a *shot tower*—and solidify as they fall.
   a. If a bearing needs $4.0\,\mathrm{s}$ to solidify enough for impact, how high must the tower be?
   b. What is the bearing's impact velocity?
10. A rock is tossed straight up from ground level with a speed of $20\,\mathrm{m/s}$. When it returns, it falls into a hole $10\,\mathrm{m}$ deep.
    a. What is the rock's speed as it hits the bottom of the hole?
    b. How long is the rock in the air, from the instant it is released until it hits the bottom of the hole?
11. A rock is dropped from the top of a tall building. The rock's displacement in the last second before it hits the ground is 45% of the entire distance it falls. How tall is the building?
12. A car traveling at $30\,\mathrm{m/s}$ runs out of gas while traveling up a 10° slope. How far up the hill will it coast before starting to roll back down?
13. A bicycle coasting at $8.0\,\mathrm{m/s}$ comes to a $5.0\,\mathrm{m}$ long, $1.0\,\mathrm{m}$ high ramp. What is the bicycle's speed as it leaves the top of the ramp?
14. A snowboarder glides down a $50\,\mathrm{m}$ long, 15° hill. She then glides horizontally for $10\,\mathrm{m}$ before reaching a 25° upward slope. Assume the show is frictionless.
    a. What is her speed at the bottom of the hill?
    b. How far can she travel up the 25° slope?
15. You're driving down the highway late one night at $20\,\mathrm{m/s}$ when a deer steps onto the road $35\,\mathrm{m}$ in front of you. Your reaction time before

stepping on the breaks is $0.50\,\mathrm{s}$, and the maximum deceleration of your car is $10\,\mathrm{m/s^2}$.

a. How much distance is between you and the deer when you come to a stop?
b. What is the maximum speed you could have before stopping and still not hit the deer?

16. The Boring Company has proposed transporting people between Los Angeles and San Francisco, a distance of $540\,\mathrm{km}$, through an underground tube from which the air has been removed to eliminate drag. Small pods carrying four passengers would accelerate at $2.5\,\mathrm{m/s^2}$ until reaching a speed of $330\,\mathrm{m/s}$. Later they would break at $1.5\,\mathrm{m/s^2}$. A launch of one pod per minute would transport 240 passengers per hour, roughly equivalent to one get plane per hour.

a. What would be the trip time in minutes from Los Angeles to San Francisco?
b. How far apart would two adjacent pods be on the constant-speed segment of the journey?

17. A hotel elevator ascends $200\,\mathrm{m}$ with a maximum speed of $5.0\,\mathrm{m/s}$. Its acceleration and deceleration both have a magnitude of $1.0\,\mathrm{m/s}$.

a. How far does the elevator move while accelerating to full speed from rest?
b. How long does it take to make the complete trip from bottom to top?